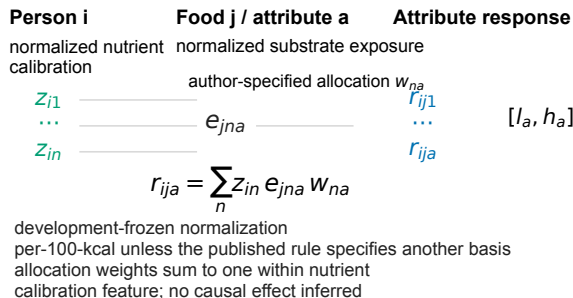


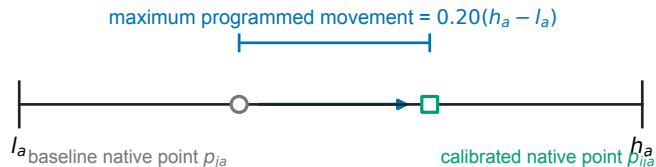
# CONCEPTUAL METHOD — NO OBSERVED OR SYNTHETIC DATA

a



b

published native point range



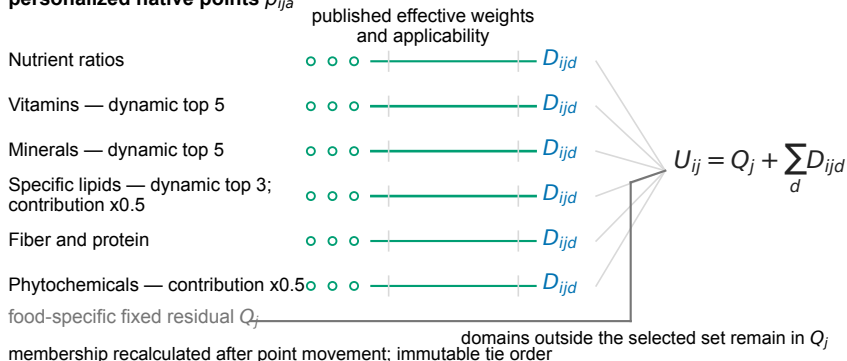
$$\text{raw\_delta}_{ija} = 0.20(h_a - l_a)\tanh(r_{ija}/2) \quad \text{primary locked mode: 20\%}$$

$$p_{ija} = \text{clip}[p_{ja} + \text{raw\_delta}_{ija}, l_a, h_a]$$

Conceptual; not to scale

c

personalized native points  $p_{ija}$



d

symbolic shared scale

Food Compass 2.0 anchor  $F_j$



allowed final deviation: -12 to +12 points

$$F_{\text{native}, ij} = \text{unscaled\_to\_FCS}(U_{ij})$$

$$S_{ij} = \text{clip}_{[1, 100]} \{ F_j + \text{clip}_{[-12, 12]} (F_{\text{native}, ij} - F_j) \}$$

score\_centering = none

shared scale retained by construction  
computational bound, not observed  
population preservation

Conceptual; not to scale